Ducted Fan

Group 32 – Emmanuel T. Abolarin , Jude Holemon, Justin Zandi , Kondwani Phiri

USC Viterbi School of Engineering, Aerospace and Mechanical Department

Background and Need

Electric ducted fans are used across aerospace and robotics because a ducted propulsor can increase mass flow and reduce tip losses, giving better static thrust efficiency than an open propeller. However, despite their widespread use, EDFs are still under characterized in static conditions, so designers lack reliable benchmark data to compare geometries or validate performance models. Non ducted propellers are known to be less efficient in static operation, but the size of the duct benefit is not well quantified without controlled testing. To close this gap, we need a repeatable, instrumented static thrust facility that can rapidly swap between open and ducted configurations, record thrust, RPM, and electrical power, and output non dimensional metrics for fair cross design comparison. This test stand also serves as a reusable lab asset that supports ongoing AME 441A research and future student cohorts.

Our goal is to build and calibrate a modular static thrust test stand for both open and ducted propellers, enabling repeatable measurements of thrust per watt and non dimensional performance metrics. We will use these data to quantify duct augmentation and demonstrate the efficiency advantage of ducted propulsors over non ducted propellers

Theory

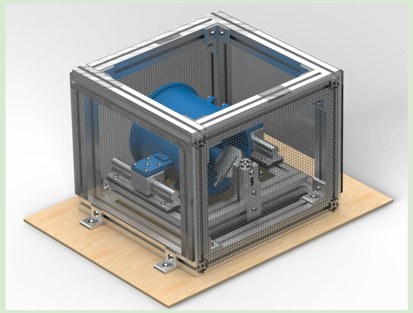
Static propeller performance is interpreted using actuator disk momentum theory for operation at near zero freestream velocity. Under steady, incompressible flow assumptions and tip Mach numbers below about 0.6, an ideal actuator disk produces thrust $T = 2\rho A v_i^2$ and ideal power $P_{i,deal} = T v_i$. These relations motivate use of a static Figure of Merit, $FoM \approx T^{3/2} / (P_{in}\sqrt{2\rho A})$, to evaluate how closely a measured propeller approaches ideal static efficiency.

To compare propeller and duct designs across different diameters and rotational speeds, results are non dimensionalized with thrust and power coefficients: $C_T = T / (\rho n^2 D^4)$ and $C_P = P_{in} / (\rho n^3 D^5)$, where $n = RPM/60$. Duct benefit is quantified by an augmentation ratio $Aug = T_{ducted} / T_{open}$, evaluated at matched RPM or matched electrical power, allowing a fair measure of duct induced thrust improvement.

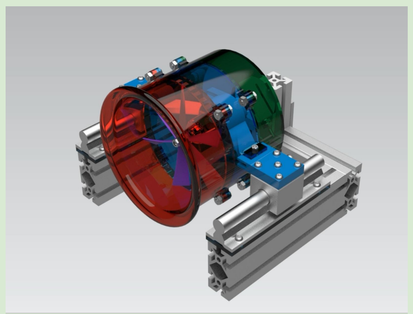
Departures from ideal scaling are examined with tip speed $V_{tip} = \pi D n$, tip Mach number $M_{tip} = V_{tip}/a$, and Reynolds number $Re = \rho V_{tip} c/\mu$. Lower Reynolds numbers tend to reduce propeller efficiency, while higher tip Mach numbers introduce additional aerodynamic losses and noise, so these parameters help explain trends across the test matrix.

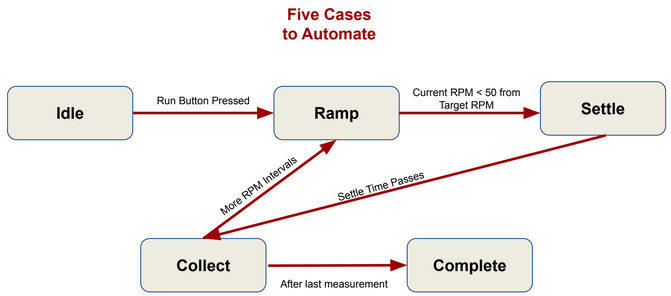
Design and Methods

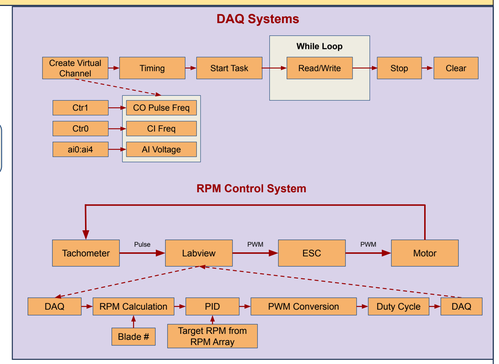
GOON Cage



Thrust Stand







Test stand design: A modular static thrust rig was built on an 80/20 aluminum base with a linear rail carriage and a quick swap motor and adapter bracket to allow rapid changes between open propeller and ducted propeller configurations. The propeller is mounted inline with a segmented duct consisting of an intake lip, center ring, and diffuser, and the thrust load path is aligned directly through an S beam load cell for accurate static thrust measurement.

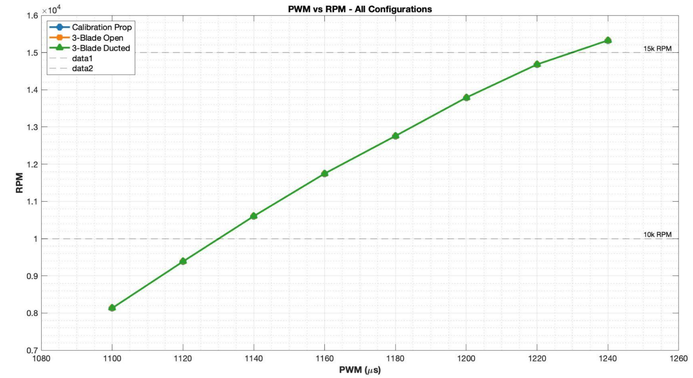
Instrumentation: The rig measures static thrust, propeller rotational speed, voltage, current, and vibration. Electrical power is computed from simultaneous voltage and current sensing, while RPM is obtained from a tachometer input. Ambient pressure and temperature are recorded to compute air density for non dimensional performance metrics.

Control and data acquisition: A LabVIEW based DAQ system controls the ESC motor command and logs all sensor channels. RPM is regulated using a PID loop with tachometer feedback. Automated test sequences run through idle, ramp, settle, collect, and complete states to ensure consistent acceleration rates and steady state sampling windows at each RPM step

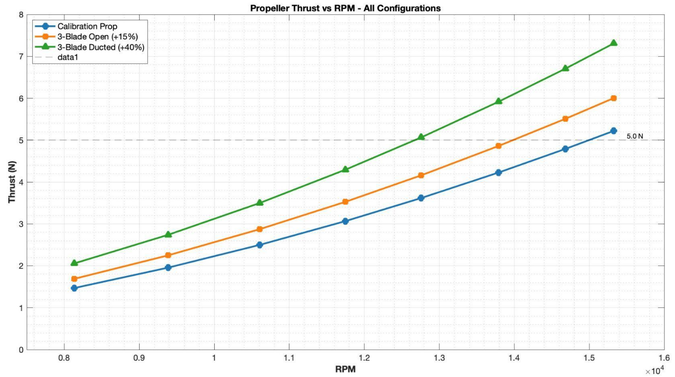
Safety procedures: All tests are conducted inside a GOON cage enclosure made from rigid 80/20 framing with steel mesh panels, a solid overhead guard, and a plywood anchor base. The enclosure includes an interlocked front service panel and an external emergency stop so operators remain outside the hazard zone during arming, runs, and shutdown.

Data

PWM vs RPM



RPM vs Thrust (N)



References